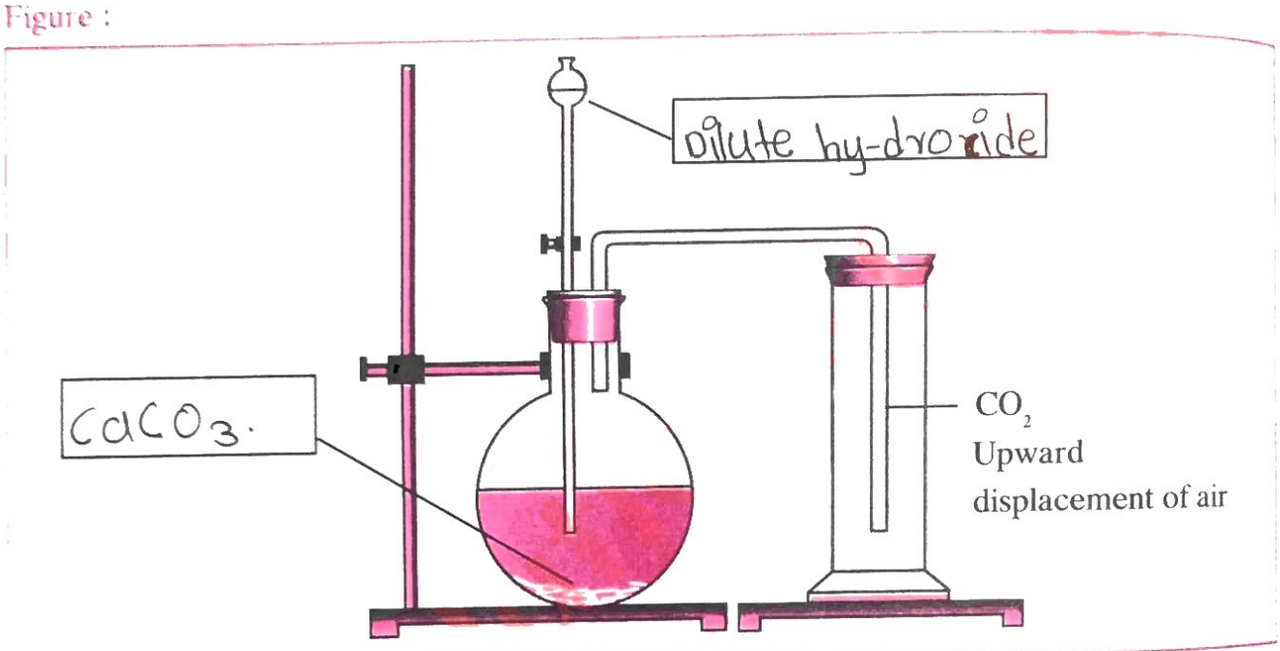
Experiment No. 12

Title: To prepare carbon dioxide in the laboratory and to study its physical and chemical properties.

Apparatus : Flat bottom flask, Rubber cork with two holes, Thistle funnel, Gas tubes, Five jars, Test tubes, Test tube stand, Stand, Candle etc.

Chemicals : Calcium carbonate (Shahabad stone pieces/ Marble pieces/Lime stone), Dilute hydrochloric acid, Freshly prepared lime water etc.

Figure :



Procedure :

1. Arrange the apparatus as shown in figure. Place $CaCO_3$ in the flat bottom flask while arranging.
2. Add dilute HCl through the funnel. Take care that the end of the funnel dips in the acid.
3. CO_2 is formed by reaction between $CaCO_3$ and dilute HCl. Collect this gas in four to five jars.
4. Use separate gas jars for the procedure 5 to 9 in the experiment.
5. Pour some lime water in the gas jar.
6. Place a burning candle in the gas jar.
7. Pour some solution of universal indicator in the gas jar and shake.
8. Pour some water in the gas jar and shake.
9. Put moist blue and red litmus in the glass jar.

Complete the equation of chemical reaction

$$CaCO_3 + 2\,HCl \rightarrow \ldots CaCl_2 \ldots + H_2O + CO_2 \uparrow$$

Explanation when $CaCO_3$ combines with 2HCl by the combination reaction $CaCl_2 + H_2O$ product is obtained and CO_2 gets out.

Observation table :

Sr.No	Procedure	Observation	Inference
1	Observe the colour of the gas formed.	colourless	co2 is the colourless
2	Smell the gas in the gas jar.	orderless	But pasing throug solution.
3	Pour some lime water in the glass jar.	lime water turn milky	It orderless
4	Place burning candle in the glass jar.	candle extinguised	It is used in fiee extingused
5	Pour some solution of universal indicator in the gas jar and shake it.	solution become red.	co2 gas is acidic in hature
6	Put moist blue and red litmus in the gas jar.	red litmus paper not effected	co2 is acidic.

Multiple choice questions

1. CO_2 turns lime water milky because .....
   A. calcium hydroxide is formed B. calcium oxide is formed
   C. calcium carbonate is formed D. calcium sulphate is formed
2. Molecular mass of CO_2 is .....
   A. 3 B. 14 C. 48 D. 44
3. .....is not a property of carbon dioxide.
   A. Combustibility B. Heavier than air C. Soluble in water D. Odourless
4. Dry ice is obtained from CO_2 at ..... °C temperature.
   A. 0 B. -56.6 C. -78.5 D. -109.3
5. The Statement .....is wrong regarding the use of CO_2.
   A. Ecofriendly drycleaning B. Removal of caffene from coffee
   C. Fire extinguisher D. Purificants in petroleum

Questions

1. What are the uses of CO_2 in everyday life?

1) carbon dioxide obtainly chemical reaction of kept under presure is used in rice extingulisher.
2) liquefied co2 is used to remom caffien from coffe.
3) liquefied co_2 is used to us solvent in dry cleanning.

2. Explain the relation between carbon dioxide and green house effect?

Ases the atomosphere such as carbon doxide do what the roof of a green house does what during the day the sun shine throught the atomosphere earth surface warms up in the sunlight

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